



EC-201: Engineering Project Management

Concepts, Definitions and Nature of Projects

Broad Contents

- What is a project?
- Why projects?
- Attributes of a project
- Characteristics of projects
- Project environment
- Project participants
- Projects and strategic planning
- Examples of projects
- Project types

2.1 What is a Project? (1 / 2)

- J. M. Juran defined that “*a project is a problem scheduled for solution.*” *Problem refers to the gap between where you are and where you want to be, with an obstacle that prevents easy movement to close the gap.*
- Projects are a group of activities that have to be performed with limited resources to yield specific objectives, in a specific time, and in a specific locality. Thus, *a project is a temporary endeavour employed to create a unique product, service or results.*
- *Project is a unique process*, consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirements, including the constraints of *time*, *cost* and *resources*.

2.1 What is a Project? (2/2)

- a. Short Range Projects:* They are completed within one year, and are focused towards achieving the tactical objectives. They are less rigorous; require less or no risk.
- b. Long Range Projects:* These projects involve higher risk and a proper feasibility analysis is essential before starting such projects. Their major impact is overlong period of time, on internal as well as external organization.

2.2 Why Projects are initiated?

Projects are initiated in the following scenarios:

- When starting a new business.
- In order to develop/ modify a product or service.
- For relocating and/or closing a facility.
- For regulatory mandate.
- For some community issues.
- In order to re-engineer the process so as to reduce complaints, eliminate errors etc.
- To introduce new equipment, tools or techniques.

2.3 Attributes of a Project

- Projects focus on a single goal as compared to a program. They have customers who are affected by the end results. Projects have to be completed within specified time frame (*completion date*), within budget (*limited resources including, people, money, machines*) and should be according to the specifications (*with a certain level of functionality and quality*).

2.4 Characteristics of Projects

- As already mentioned projects are temporary with a definite beginning and a definite end.
- They also have temporary opportunities and temporary teams.
- Projects are terminated when the objectives are achieved, or conversely, if the objectives cannot be met.
- Most of the projects last for several years. However, they have a finite duration.
- They involve multiple resources (human and non-human) and require close coordination.
- They are composed of interdependent activities.
- At the end of the project, a unique product, service or result is created. Some degree of customization is also a characteristic of projects.
- Projects encompass complex activities that are not simple, and may require repetitive acts.
- They also include some connected activities. Some order and sequence is required in project activities. The output from one activity is an input to another.
- Project Management lives in the world of conflict. The management has to compete with functional departments for “resources and personnel”.
- There exists a constant conflict for project resources and for leadership roles in solving project problems.
- In every project, clients want changes, and the parent organization aims at maximization of profits.
- There can be two bosses at a time and that too with different priorities and objectives.

2.5 Project Environment (1/2)

All projects are planned and implemented in a social, economic, environmental, political and international context.

- a. Cultural and Social Environment:* is that how a project affects the people and how they affect the project. This requires understanding of economic, demographic, ethical, ethnic, religious and cultural sensitivity issues.
- b. International and Political Environment:* refers to the knowledge of international, national, regional or local laws and customs, time zone differences, teleconferencing facilities, level of use of technology, national holidays, travel means and logistic requirements.
- c. Physical Environment:* is the knowledge about local ecology and physical geography that could affect the project, or be affected by the project.

2.5 Project Environment (2/2)

THE PROJECT ENVIRONMENT



Figure 2.1: Project Environment

2.6 Project Participants (1/5)

- a. **Stakeholders:** are the ones *who have a share, or an interest in an enterprise*. Stakeholders in a company may include *shareholders, directors, management, suppliers, government, employees, customers, and the community*. Stakeholders are influenced by the outcomes and objectives. They have varying level of responsibility and authority. Thus, *they should not be ignored*. A project manager should try to manage and fulfil the expectations of the stakeholders. *The project management team must identify the stakeholders, determine their requirements and expectations*. Stakeholders may have a positive or negative influence on a project.

2.6 Project Participants (2/5)

- a. *Stakeholders:(continued...) Positive stakeholders are those who would normally benefit from a successful outcome from the project, while negative stakeholders are those who see negative outcomes from the project's success. For example, business leaders from a community that will benefit from an industrial expansion project may be positive stakeholders because they see economic benefit to the community from the project's success. Conversely, environmental groups could be negative stakeholders if they view the project as doing harm to the environment.*

2.6 Project Participants (3/5)

- b. Key Stakeholders:** Key stakeholders include the following:
- i. Project Manager-** The person, who is responsible for managing the project.
 - ii. Customers, End Users-** The person or organization that will use the project's product.
 - iii. Project Team Members-** The group that is performing the work of the project. It includes the members who are directly involved in the project activities.
 - iv. Sponsors-** The person or group that provides financial resources, in cash, or kind, for the project.
 - v. Influencers-** People or groups that are not directly related to the use of the project's product.
 - vi. Project Management Organization-** If project exists in performing organization, the Project Management Organization can be a stakeholder if it has direct responsibility for the outcomes of the project.

2.6 Project Participants (4/5)

- c. **Project Stakeholders:** In addition to the key stakeholders, there are many different names and categories of project stakeholders, influencing internal or external, owners and investors, sellers and contractors, team members and their families, government agencies, media outlets, individual citizens and society-at-large. Project Managers must manage stakeholder expectations, which can be difficult because stakeholders often have very different or conflicting objectives. For example:
- The *vice president of research at an electronics* firm may define new product success as state-of-the-art technology, the *vice president of manufacturing* may define it as world-class practices, and the *vice president of marketing* may be primarily concerned with the number of new features.

2.6 Project Participants (4/5)

c. *Project Stakeholders: (continued...)*

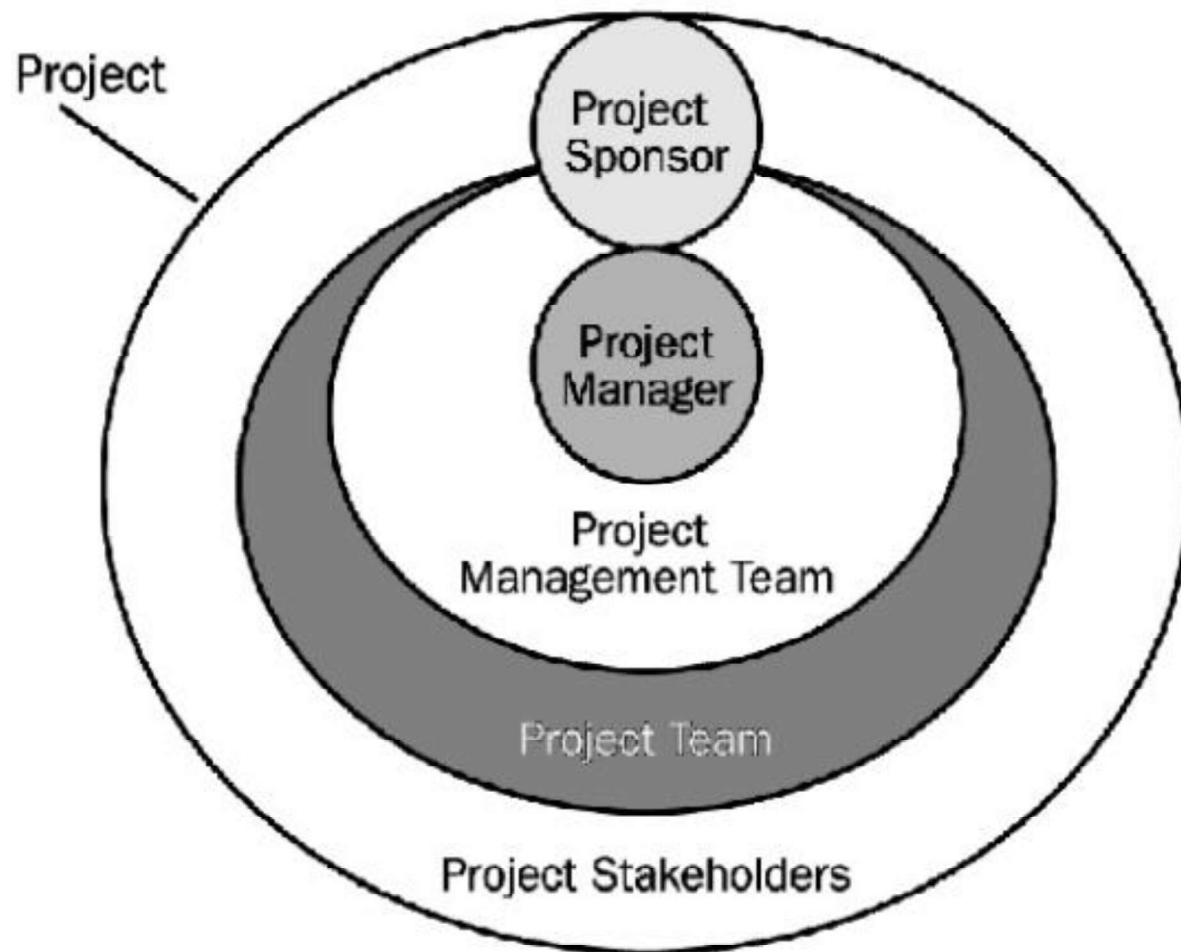


Figure 2.2: Stakeholders and Projects

2.6 Project Participants (5/5)

c. *Project Stakeholders: (continued...)*

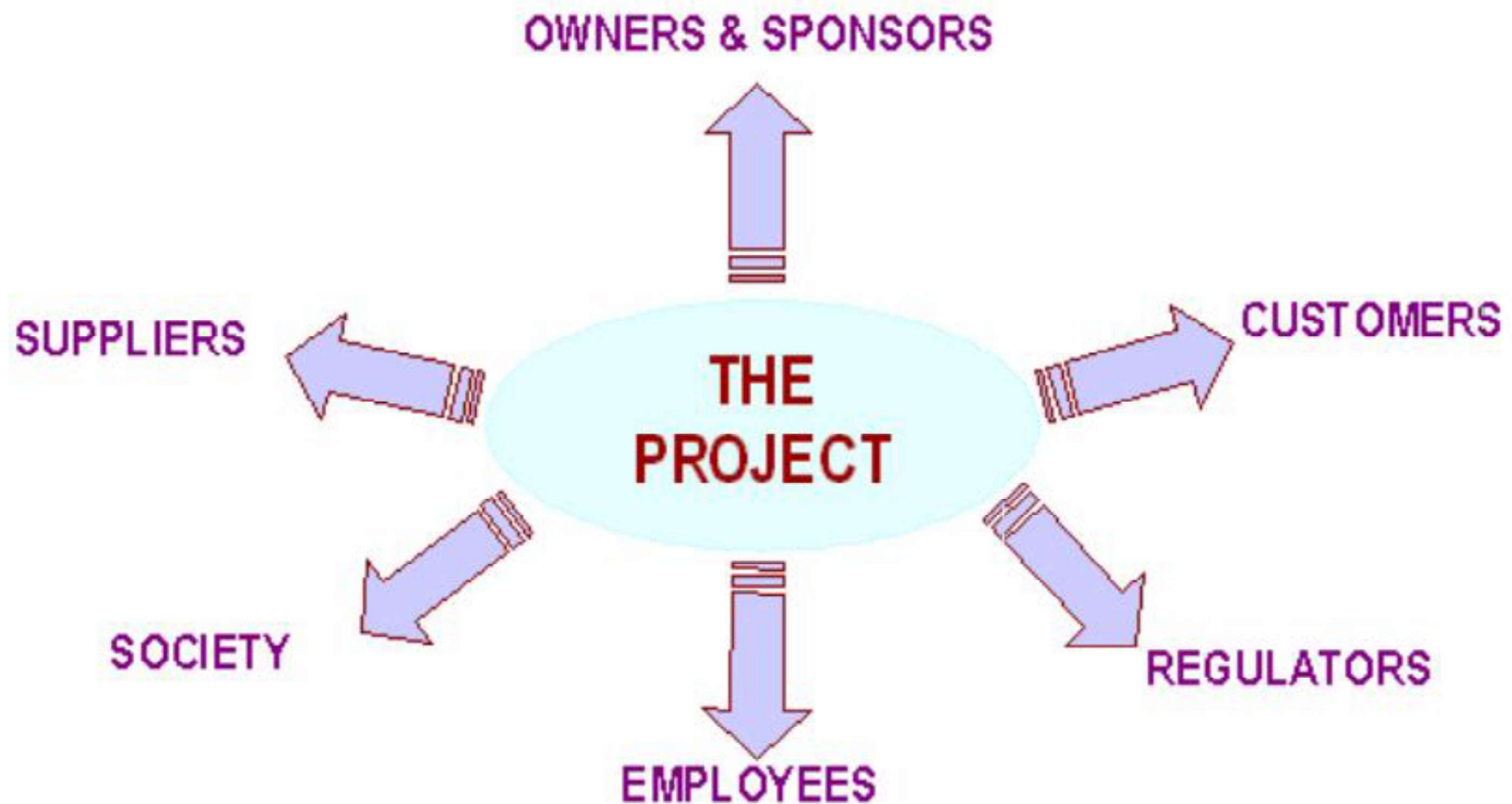


Figure 2.3: Relevant Stakeholders

2.7 Projects and Strategic Planning

Projects are the means of achieving organization's strategic plans. Following are the strategic considerations that have to be kept in mind while planning for projects:

- The market demand (e.g. a new refinery).
- Organizational needs (e.g. a university offers new courses for revenue generation).
- Customer's requests (e.g. an Internet Service Provider ISP provider lunches DSL).
- Technological demand (e.g. new video games, new cell phones with advance features).
- Legal requirements (e.g. child labor control project, toxic waste disposal center).

2.8 Sub Projects

Projects are frequently divided into more manageable components or sub projects. Individual sub projects are also a project and are managed as such.

2.9 The Triple Constraint of Project Management (1/2)

Meeting stakeholder needs and expectations *involves balancing competing demands among cost, quality, scope, and time* i.e., $Q = f(T, C, S)$.

- Where Q is Quality, C is Cost, S is Scope and T is Time.
- Project quality is affected by balancing these three (*time, cost and Scope*) factors.
- Projects fail when:
 - ☐ Estimates are faulty
 - ☐ Time, talent and resources are insufficient or incorrectly applied

2.9 The Triple Constraint of Project Management (2/2)

Figure 2.4 is a pictorial representation of project management. The objective of the figure is to show that *project management is designed to manage or control company resources on a given activity, within time, within cost, and within performance expectations.*



Figure 2.4: Overview of Project Management

2.10 Example of Projects

Examples of Projects:

- Hosting a web site of your department.
- Executing an environmental clean-up of a contaminated site.
- Holding a University alumni reunion.
- Developing a new product or service.
- Developing or acquiring a new or modified information system, etc.

2.11 Operations and Projects (1/2)

Operations are ongoing and repetitive activities conducted by the staff. Some of these include:

- Financial management and control
- Continuous manufacturing
- Product distribution

Projects are temporary and unique, and are performed by teams that have:

- Clearly defined team and individual roles
- Open and effective communication systems
- Visible rewards for good performance, and have constant pressure to improve poor performance

2.11 Operations and Projects (2/2)

Common Characteristics between operations and projects are as follows:

- They are both performed by people
- They are constrained by limited resources
- Both are planned, executed, and controlled

2.12 Project Types (1/4)

		Project End Requirements	
Project Methods		Well Defined	Poorly Defined
	Well Defined	Type I Construction	Type III Software
	Poorly Defined	Type II Product Development, Space	Type IV OD, Vision, Training Assessment

Figure 2.5: Project Types

2.12 Project Types (2/4)

Refer to Figure 2.5 previous slide we have:

- ***Type I Projects – Large Engineering Projects:***
They have well defined project methods and end project requirements, such as construction projects.
- ***Type II Projects – Product Development Projects, Early Space Projects:***
They have poorly defined project methods but have well defined project end requirements.
- ***Type III Projects – Software Development Projects:***
In these, the shape of end product proceeds. They have well defined project methods, but poorly defined project end requirements.
- ***Type IV Projects – Organizational Development Projects, Vision Definition, Assessment of Impact of Trainings:***
They have both poorly defined project methods as well as project end requirements.